

INDUSTRIAL TRAINING REPORT

At

BUILDTECH INDIA

(BAZAR NO. 2, FIROZPUR)

**A PROJECT REPORT
SUBMITTED IN PARTIAL FULFILMENT OF
THE REQUIREMENT FOR THE AWARD OF THE DEGREE OF**

**BACHELOR OF TECHNOLOGY IN
CIVIL ENGINEERING
BY**

ASHISH KUMAR (1908208)

SUBMITTED TO

ER. GURPREET SINGH
(TPI OF CIVIL ENGINEERING DEPARTMENT)



**DEPARTMENT OF CIVIL ENGINEERING
SHAHEED BHAGAT SINGH STATE UNIVERSITY
FEROZEPUR, 152004 (PUNJAB)**

ACKNOWLEDGEMENT

I express my deep gratitude to Authorities of “**Military Engineer Services**” for their valuable suggestions, help and guidance throughout the period of training. I am also thankful to the other team members for their support and guidance.

I am also thankful to site engineer **MR. RAJINDER** and contractor **MR. VIKRANT SALUJA** who were there with their knowledge and help whenever I felt need of anything.

I am also extremely thankful to **DR. DAPINDER DEEP SINGH** Head of Department of Civil Engineering branch, Shaheed Bhagat Singh State University, Firozpur, for his constant guidance and motivation.

I am also thankful to **ER. GURPREET SINGH** (Training and Placement In-charge) for his trust and assigning the required permission.

DECLARATION

I hereby confirm that, I **Ashish Kumar** student of **B.Tech (Civil Engineering) Roll no. 1908208** has undergone industrial training of 3 months (January-March) in “**BuildTech India Pvt Ltd**” organization as per syllabus of 8th semester required for the award of degree of B.Tech (Civil Engineering) from Shaheed Bhagat Singh State University (formerly known as SBS state technical campus).

I also declare that this report is the result of my own effort and that the same has not been submitted to any other university, institution for the award of any degree or diploma.

SIGNATURE



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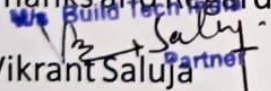
Ref. No. VUE/Experience / 1498

Date 07-01-2023

To
 The TPI
 Department of Civil Engineering
 SBSSU
 Ferozepur, 152004

Respected Sir

It is pleasure to confirm that the student named Ashish Kumar, Roll No. 1908208, student of B.tech Civil Engineering from Shaheed Bhagat Singh State University, will work under our firm as an Industrial training period starting from 1st January 2023 to 31st March 2023. The management will view his performance continuously during the period of his training.

Thanks and Regards

 Vikrant Saluja
 (Govt. Contractor)





Shaheed Bhagat Singh State University
Moga Road (NH-95), Ferozepur-152004 (Punjab)

(An Engineering Institute of Punjab Government)
Industrial/Software Training Evaluation Format
(Mid-Semester)

Student Name : Ashish Kumar

University Roll No. : 1908208

Organization Name & Address : Build tech India

Head office : # 102, Bazar no.2, FZR cant
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S. No.	Key Aspects	Total Marks	Marks Awarded
1.	Attendance	05	5
2.	Performance (Work done /simulation/hardware/project developed)	15	15
3.	Daily Dairy	05	3
4.	Total	25	23

Ms Build Tech India
Partner

Signature : [Signature]
Name : Mr. Vikrant Saluja
Phone : 9814787721

(Training Incharge from Industry)

Signature : _____
Name : _____
Phone : _____

(Faculty who visited Industry)



Shaheed Bhagat Singh State University
Moga Road (NH-95), Ferozepur-152004 (Punjab)

(An Engineering Institute of Punjab Government)
Industrial/Software Training Evaluation Format
(End-Semester)

Student Name : Ashish Kumar
University Roll No. : 1908208
Organization Name & Address : Buildtech India
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Branch office: College road, Pathankot
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S. No.	Key Aspects	Total Marks	Marks Awarded
1.	Attendance	05	5
2.	Performance (Work done /simulation/hardware/project developed)	15	15
3.	Daily Dairy	05	4
4.	Total	25	24

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Ref. No. VUE/Experience/1519

Date 16-04-2023

TO WHOMSOEVER IT MAY CONCERN

This is to certify that Ashish Kumar, Roll No. 1908208 of Shaheed Bhagat Singh State University (formerly known as SBSSTC) has undergone industrial training in our organization from 01 Jan 2023 to 31 March 2023. He was trained under various activities at the construction site.

During his training we found him to be sincere & hardworking.

We wish him best for his future endeavors.

Buildtech India

We Build Tech India
Vikrant Saluja
Partner

(Government Contractor)



TABLE OF CONTENT

Sr. no.		Title	Page no.
1.		INTRODUCTION	9
2.		PROJECT DETAIL	11
3.		GEOGRAPHICAL LOCATION OF SITE	11
4.		PLAN OF THE BUILDING	12
5.		SITE VIEW	13
6.		MATERIAL USED	14
	6.1	CEMENT	14
	6.2	AGGREGATE	14
	6.3	WATER	15
	6.4	STEEL	15
	6.5	SAND	16
	6.7	SOIL	16
	6.8	BRICK	16
7.		EQUIPMENT AND TOOLS USED	19
	7.1	CONCRETE MIXER	19
	7.2	HAND TAMPER	19
	7.3	VIBRATOR	20
	7.4	EXCAVATOR MACHINE	21
	7.5	TROWEL	21
	7.6	PLUMB BOB	22
	7.7	MEASURING TAPE	22
	7.8	HOE	22
	7.9	HEAD PAN	23
	7.10	CONCRETE PLANT	23
	7.11	DUMPY LEVEL	24
	7.12	TRACTOR	24
8.		METHODOLOGY OF WORK	25
	8.1	SITE PREPARATION OR LEVELING	25
	8.2	SITE SURVEY AND LAYOUT	25
	8.3	EXCAVATION	26
	8.4	PCC LAYING	27
	8.5	BAR BENDING	27
	8.6	SHUTTERING	28
	8.7	FOUNDATION	29
	8.8	COLUMN	30
	8.9	PLINTH BEAM	32
	8.10	BACKFILLING	34
	8.11	SUPERSTRUCTURE	34
	8.12	MASONARY WORK	35
9.		CONCLUSION	36

INTRODUCTION

The Military Engineer Services (MES), which was raised in 1923, is one of the pillars of Corps of Engineers of the Indian Army which provides rear line engineering support to the Armed Forces.

It is one of largest construction and maintenance agencies in India. It has a large number of units and sub units spread across the entire country to provide engineering support to various formations of Army, Air Force, Navy, Ordnance Factories, KV Sangathan, Border Road Organisation and DRDO. MES is a military organisation but has both Army and Civilian component of officers and other subordinate staff. While execution of all construction works is through contracts, execution of all maintenance services are both through contracts as well as departmentally employed labour (DEL).

The role of MES is dual i.e. to render both engineering advice and also to execute the works .The Military Engineering Services are responsible for the design, construction and maintenance of all works, buildings, airfields, dock installations, etc. along with necessary services such as military roads, bulk water and electricity supply, drainage, refrigeration and furniture, required by the Army, Navy and Air Force in India.

The MES functions under the overall control of the Engineer-in-Chief, who is the adviser to the Ministry of Defence and the three Services on operational and peace time construction engineering MES is structured to design works which are executed through contracts under the supervision of officers and staff consisting of both civilians and combatants from the Corps of Engineers.

It has an integral multi-disciplinary team of architects, civil, electrical and mechanical engineers, structural designers, quantity surveyors and contract specialists for planning, designing and supervision of works.

MES (Military Engineer Services)

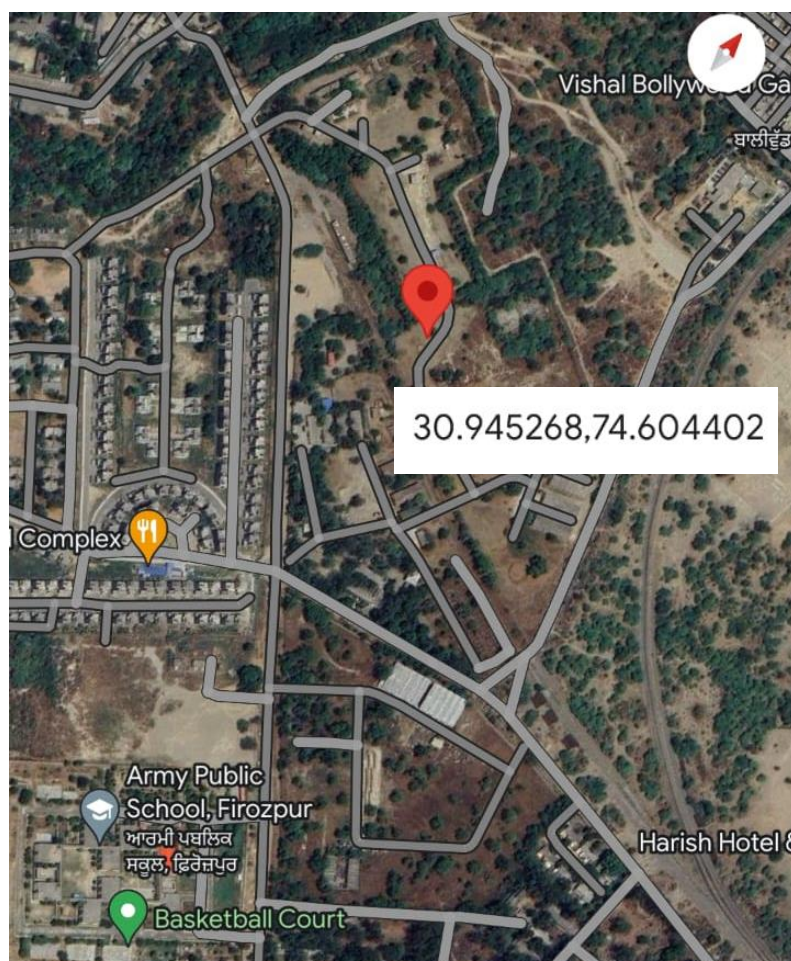


Formation	26 September 1923
Type	Government Organization
Location	New Delhi
Region	India
Director General	Abhay Bhatnagar, IDSE
Engineer-in-chief	Lt Gen. Arvind Walia
Website	mes.gov.in

PROJECT DETAIL

NAME OF THE PROJECT	ARMY GOODS STORAGE SHEADS
LOCATION	INSIDE ARMY SUPPLY DEPOT, NEAR ARMY PUBLIC SCHOOL, FEROZEPUR CANTT.
COST ESTIMATION	10.5 CRORES
TIME ESTIMATION	18-24 MONTHS
CONTRACTOR	BUILDTECH INDIA (MR. VIKRANK SALUJA)

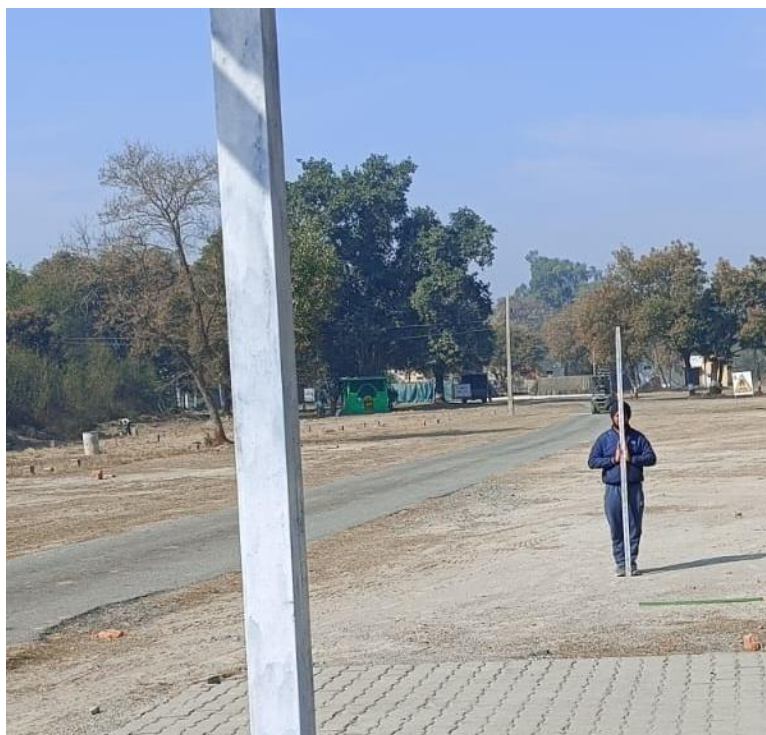
GEOGRAPHICAL LOCATION OF SITE



PLAN OF THE BUILDING



SITE VIEW



MATERIAL USED

CEMENT :-

The function of cement is to combine with water and to form cement paste. This paste sets and binds the aggregates together into a stone like hard mass and thus provide strength & durability. Brand of the cement used is Ambuja cement. Quantity used of the cement was in ratio and was used in various proportions such as 1:4:6 or 1:1.5:3 or 1:6. Cement unloading was done by labour and was kept under metallic roof protected from rain and direct sunlight.



AGGREGATE :-

Aggregates are small pieces of broken stones in-irregular size and shapes.



✚ TYPES OF AGGREGATES :

- ❖ Fine aggregates
- ❖ Coarse aggregates

Fine aggregates are small size filler materials in construction. Fine aggregates are the particles that pass through 4.75 mm sieve and retain on 0.075 mm. Sand, surki, stone screenings, burnt clays, cinders, fly ash, etc. are used as fine aggregate in concrete. Fine aggregates are used for casting mixed with sand.

Coarse aggregates are larger size filler materials in construction. coarse aggregates are the particles that retain on 4.75 mm sieve. For PCC under column footing coarse aggregate used was actually stone ballast which was about 30 to 50 in millimeters.

Brick chips (broken bricks), stone chips (broken stones), gravels, pebbles, clinkers, cinders etc. are used as coarse aggregate in concrete.

✚ WATER :-

It is an important ingredient of concrete because it combines with cement and forms a binding paste. The paste thus formed fills up the voids of the sand and coarse aggregate bringing them into close adhesion.

✚ STEEL :-

Though plain cement concrete has high compressive strength and its tensile strength is relatively low. So it is reinforced by placing steel bars in the tensile zone of the concrete beam so that the compressive stress is carried by concrete and tensile stress is carried by steel reinforcing bars. Steel used was of different sizes such as 8mm, 10mm, 16mm and 20mm.

Steel was used in various prospects like in footing, in column binding and other parts of building too.



✚ SAND :-

Sand is a must adding material in the concrete and mortar as it stops the quick binding and drying of the cement paste. In other words it helps keeping the mix wet until few hours.

○ These are of three types:

- Coarse sand
- Medium sand
- Fine sand

▪ Coarse sand:-

It is one which contains 90% of particles of size greater than 0.6 mm and less than 2 mm.

▪ Medium sand:-

It is one, which contains 90 % of particles of particles size greater than 0.2 mm and less than 0.6 mm.

▪ Fine sand:-

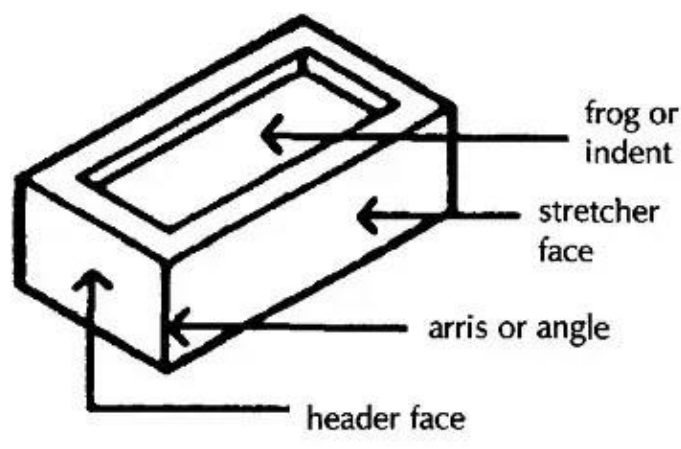
It is one, which contains 90% of particles of size greater than 0.06mm and less than 0.2 mm.

Proper selection of sand is critical in the durability and performance of concrete mixture.

✚ SOIL :-

The soil excavated from the site is used for backfilling process after construction of the columns and footings. For both processes excavator is used.

✚ BRICK :-

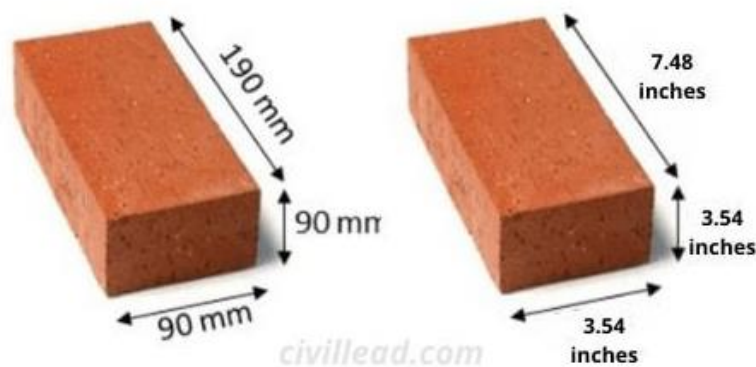


A brick is a type of construction material used to build walls, pavements and other elements in masonry construction.

Bricks are laid in courses and numerous patterns known as bonds, collectively known as brickwork, and may be laid in various kinds of mortar to hold the bricks together to make a durable structure.



- **Standard size of brick :- (19 x 9 x 9)cm**



- **Brick Bond :-**

A systematic arrangement of a course of bricks in brick masonry, by which the continuous vertical joints are formed, is called a Bond.

The most commonly used types of bonds in brickmasonry are:-

i. Stretcher bond :-



ii. Header bond:-



iii. English bond:-



iv. Flemish bond:-



EQUIPMENT AND TOOLS USED

CONCRETE MIXER :-

This is a power mechanically operated machine which is used to mix the concrete.

It consist of a hollow cylindrical part with inner side wings. In which cement, sand, aggregate and water are mixed properly.

Concrete mixer consists of a large barrel or drum with an open mouth on one end, affixed to a rotating shaft with an engine to turn it. As the drum turns, the concrete inside mixes together. When it's ready, the barrel can be tilted to pour into a mold or wheelbarrow, or the mixture can be shoveled out.



HAND TAMPER :-

Tampers are manually operated or power compaction tools. They increase the density and levelness of soil, gravel, sand and other materials by applying force to the material's surface. This force can be delivered by hand or via a gas-burning engine.

- Most tampers work by repeatedly striking the surface being compacted, flattening.
- Packing it down to increase density. A few work by rolling a heavy weight over the surface.



VIBRATOR :-

Concrete vibrators are used for construction based applications. They come in several different shapes and sizes. It is used for concrete compaction mainly in constructing roads, railways, and buildings. It consolidates freshly poured concrete to remove the trapped air and excess water.

It is to settle the concrete firmly in the place of form-work.

This is done to do the proper consolidation of concrete and avoid product defects. These vibrators are designed with the frequencies of vibration ranging from 2800 to 15000 rpm.



EXCAVATOR MACHINE :-

The excavator machine i.e. JCB is used for excavating the soil . By using this machine, the excavation can be done easily , it takes less time.



TROWEL :-

Trowel is used to lift and apply the cement mortar in small quantities. It is made of steel and wooden handle is provided for holding. The ends of trowel may be pointed or bull nosed.



PLUMB BOB :-

Plumb bob is used to check the verticality of structures. It contains a solid metal bob connected to the end of a thread. It is also used in surveying to level the instrument position.



MEASURING TAPE :-

Measuring tape is used to check the thickness, length, widths of masonry walls, foundation beds, excavated trenches etc.

HOE :-

Hoe is also used to excavate the soil by hand but in this case the metal plate is provided with acute angle to the wooden handle.



HEAD PAN :-

Head pan is made of iron which is used to lift the excavated soil or cement or concrete to the working site etc. it is more commonly used in construction sites.



CONCRETE PLANT :-

Concrete plant is the machine which is used to make concrete mix in excess amount. The two tanks are loaded with sand and aggregate. Water is provided in bulk amount through a pipeline provided from a storage. Cement is added at the head through motor mechanism.



DUMPY LEVEL :-

Dumpy level is a surveying gadget which is used to check level of the ground at various points at one or more stations. It has a lens, zooming scale, rotator and leveling bubble. It is tightened on the tripod stand adjusted according the viewer's height.



TRACTOR :-

A tractor is an engineering vehicle specifically designed to deliver a high tractive efforts at slow speed for purpose of hauling a trailer or machinery such as that used in agriculture, mining and construction.



METHODOLOGY OF WORK

There are some methods for constructions step by step :-

- SITE PREPARATION OR LEVELING
- SITE SURVEY AND LAYOUT
- EXCAVATION
- PCC LYING
- BAR BINDING
- SHUTTERING
- FOUNDATION
- COLUMN
- PLINTH BEAM
- BACKFILLING
- SUPERSTRUCTURE
- MASONRY WORK
- THE LINTEL ABOVE DOORS AND WINDOWS
- DOORS AND WINDOW FRAMES
- PLASTER WORK AND FINISHING

SITE PREPARATION OR LEVELING :-

Before starting the work on-site, it should be adequately cleaned, including removing grass, debris, and ground leveling.

SITE SURVEY AND LAYOUT :-

After cleaning the construction site, the next step is a site survey and layout. The layout survey is the process of interpreting the building plan and marking the position of the proposed

structure on the ground surface using survey equipment such as the total station so that excavation for the foundation can be carried out. In survey basic leveling was done with the help of Dumpy level. However, graph sheet and excel sheet was prepared with proper reading by Height of the Instrument method.

During layout process, column marking was done by reference marking which was marked on the basis of brick binding, kept at the distance of 3 meters away from the column positioning.



✚ EXCAVATION :-

After the layout process, Excavation was initiated for the setting up the foundation and column. Ground excavated was 2m deep from the benchmark. Overall excavation was done by the excavator however further to make the ground level equal, manually excavation was also done with the help of hoe.



PCC LAYING :-

Then a layer of PCC (plain cement concrete) was laid in the excavated pits and trenches. The concrete ratio of PCC was 1:4:8. The aggregate used in PCC was actually stone ballast which was in 40 to 50 in millimeters. Later column marking was marked on the thin layer of mortar.



BAR BINDING :-

TMT steel bars of dimensions 8mm, 10mm, 12mm, 16mm and 20mm were used. Stirrups or rings were made of 8mm. Folding of stirrups was at an angle of 135degrees and was continued as per 2D and 4D whichever required.



SHUTTERING :-

Shuttering is basically providing profile to the structure which says how our structure will appear. For foundation shuttering was 12 inches high and was put in rectangular pattern and for column. Metallic plates were screwed around the reinforcement. Shuttering is done at various steps such as foundation, columns, beams, roof casting, etc.

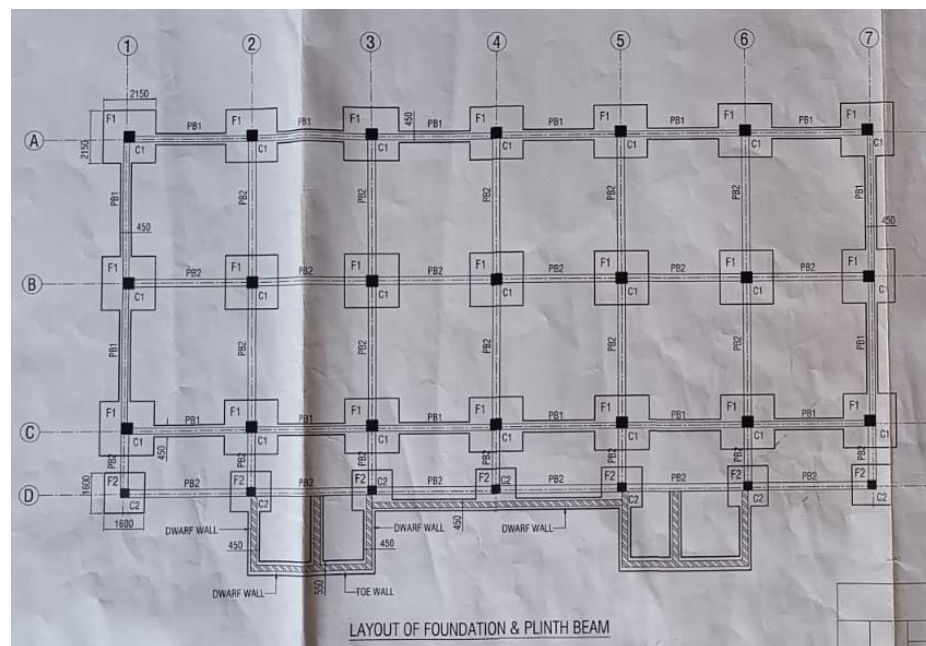


✚ FOUNDATION :-

Foundation is the lowest artificially prepared part of the structure, which is in direct contact with the ground and transmits the load of the superstructure to the ground.



SCHEDULE OF FOOTINGS :-



SCHEDULE OF FOOTINGS						
(REFER TYPICAL DETAIL OF FOOTINGS)						
FOOTING NO.	FOOTING SIZE			BOTTOM REINFORCEMENT		REMARKS
	OVER ALL	HEIGHT		ALONG SHORT SPAN	ALONG LONG SPAN	
	L X B	d	D			
F1	2150 x 2150	250	500	#12 @ 150	#12 @ 150	
F2	1600 x 1600	150	400	#12 @ 150	#12 @ 150	

Trapezoidal found was laid with proper reinforcement. Steel bars used in foundation were of size 12mm.

Size of hall foundation: (2150 x 2150) mm

Size of verandah foundation: (1600 x 1600) mm

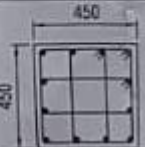
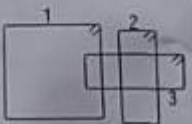
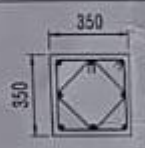
Later, Brick work was done up to bottom of the plinth level.

COLUMN :-

Column concrete ratio was 1:2:4. All work was combination of nominal mix and design mix. All columns were constructed along with the foundation and were continued at regular interval and each column was divided in two spans A and B.



SCHEDULE OF COLUMN:-

SCHEDULE OF COLUMNS						
COL. NO.	SIZE	LONGITUDINAL	SET OF TIES		SECTIONS	SHAPE AND SET OF TIES
			ZONE A	ZONE B		
C1	450x450	12-#16	10# @ 75 c/c	10# @ 150 c/c		
C2	350x350	8-#16	10# @ 75 c/c	10# @ 150 c/c		

Dimension of hall columns: 350 x 350 mm

Dimension of verandah columns: 450 x 450 mm

Column construction was continued after backfilling process and plinth beam construction.

PLINTH BEAM :-

Now bar binding and shuttering work for the plinth beam is completed. Then it is poured with concrete.

A plinth beam is a component of the superstructure located between the ground level to assist in transmitting the load of the structure to the foundation.





SCHEDULE OF PLINTH BEAM:-

SCHEDULE OF BEAMS													
S.NO.	BEAM NO.	SIZE		REINFORCEMENT								SPACER BAR (@ 1000 C/C)	REMARKS
		WIDTH (R)	DEPTH (D)	BARS THROUGH		EXTRA BARS		STIRRUPS					
				BOTTOM BAR	TOP BARS	EXTRA BOTTOM AT MID SPAN	EXTRA TOP AT SUPPORTS	X1		X3			
								DIA	SPACING C/C (P1)	DIA	SPACING C/C (P2)		
PLINTH BEAM													
1.	PB1	300	450	3- #16	3-#16	-	2-#12	# 8	100	# 8	150	#25	
2.	PB2	300	450	2- #16 + 1-#12	3-#12	-	2-#12	# 8	100	# 8	150	#25	

Size of plinth beam:

WIDTH :- 300mm DEPTH :- 450mm

Steel bar used :-

PB 1 :- 3 no. of 16 mm at bottom and 3 no. of 16mm at top.

PB 2 :- 2 no. of 16mm + 1 of 12mm at bottom and 3 no. of 12mm at top.

BACKFILLING :-

It is the method of restoring or reusing soil removed during excavation. Backfilling helps in strengthening the foundation and supporting it and other structural members.



SUPERSTRUCTURE :-

The structure above the plinth level is known as a superstructure. It consists of all parts of building like walls, columns, beams and roof. Basically, Superstructure involves construction of beams up to roof and completing load distribution.



MASONRY WORK :-

After all the above discussed work, Masonry work is executed. Bricks used were of fly ash and mortar of 1:6 was used in about 10 to 12mm and english bond is used.



CONCLUSION

The main aim of studies within this project was to investigate how a structure is constructed within its desired properties. We get knowledge about the basic & advanced techniques of building construction as well as saw the challenges which a civil engineer has to face during construction. i.e. labour problems, cost management, environmental challenges etc.

We cleared our many doubts regarding building construction. We have been studying some mechanic or electro-mechanic machinery. Overall it must be said that the construction methods and quality control on a construction.

Project needs a very good coordination and large quantities of man power, equipment and funds. During the period of Three months all the company staff helped us a lot to provide all the information about any query. So we are grateful to all the staff of **BuildTech India Pvt. Ltd.** as well as we are so thankful to the workers and labourers.

Building Construction is a significant, lifelong investment. As a result, the real estate industry is one of the prime investments ever made.

The experiences I gained through this training program will be a strong foundation to my career as an engineer. I hope that the university would continue to conduct such opportunities for us even more

